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Autonomous navigation of tethered drone strings in cluttered environments through real-time obstacle detection and geometric tether control

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ABSTRACT

This paper aims to fill the gap by identifying and critiquing the fundamental limitation of current tethered-drone systems by introducing a detailed framework of autonomous navigation in a highly obstacle-filled setting. The concept of the serially connected quadrotor is further stretched to allow real time collision avoidance and intact tether and system stability in GPS-denied cluttered environments. The suggested solution would combine vision-based sensing at the drones to detect obstacles and map the environment at a 3D scale. The hierarchical control structure is an integration of three components wherein geometric tether attitude control is involved, multi-objective model predictive control (MPC) path planning, and adaptive tether slack management in the case of emergency maneuvers. Three new contributions are made. A dynamic tether reconfiguration algorithm is first computed, which is an algorithm that calculates intermediate waypoints that remodel the tether geometry around obstacles without compromising tautness and stability. Second, the distributed artificial potential field technique, in which inter-drone communication is performed, eliminates drone-obstacle and tether-obstacle collision. Third, adaptive emergency protocol will allow an ability to avoid quickly using controlled temporary slack with assured system recovery. The MPC formulation is a combination of trajectory tracking, obstacle clearance, tether tension maintenance and control effort which subjects safety constraints to thrust limit, tension limit and clearance distance minimum. Simulated experiments show that 10 tethered drones can navigate successfully in a complex obstacle field (0.8 m average clearance and safe tension range (3–8 N)). In the proposed method, the risk of collision is reduced by 73%, and the chances of mission are improved by 42%, based on an average MPC solve time of 50.8 ms per control cycle and a deadline-aware fallback mechanism, which ensures that all control output are produced at every sampling instant with a bounded-latency. The given framework allows tethered drone delivery in the limited industrial areas, search and rescue use in cities, and infrastructure inspection as well as GPS-deprived areas that need autonomous obstacle avoidance functions.

1. Introduction

UAVs have transformed many spheres of application such as infrastructure surveying, search and rescue, precision farming, and the environment, to name a few [1,2]. Quadrotor systems have become very popular among other UAV configurations because they are very manoeuvrable, they can take off and land vertically, and they have a relatively simple mechanical design [3]. Nevertheless, the traditional free-flying quadrotors have two inherent constraints, which limit their deployment in operations to a considerable degree. First, battery power

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in the aircraft limits flight duration to between 20–40 min, which is very limiting to missions and operational range [4]. Second, localization that relies on Global Positioning System (GPS) signals makes these platforms unreliable in GPS-denied or degraded environments like indoor facilities, urban canyons, underground facilities and electromagnetically contested areas [5,6]. Tethered drone systems offer an interesting remedy to these root limitations because they maintain a physical relationship between the aerial platform and a ground station via a multifunctional tether [7]. The position can be estimated using the mechanical connection to the tether, which allows for reconstruction of the spatial configuration of each drone in the string by measuring the length of each tether segment, the angle between the joints, and the tension applied at known ground station anchor points, without the need for GPS [8]. Unlike other GPS denied positioning techniques like visual slam and ultra-wideband (UWB) ranging, the geometric positioning approach is inherent to the physical tether topology and does not suffer from drift. It has also been shown that the idea of single tethered drones has been widely explored and has shown successes in persistent surveillance, communication relay, and aerial manipulation activities [9,10]. Nonetheless, single tethered systems do not have any inherent capability to maneuver around obstacles or to reach places without a clear line-of-sight to the ground station.

The idea of serially linked tethered drones, in which several quadrotors are connected in a string layout, is an enormous enhancement to the solitary tethered systems [11]. Such an architecture could be thought of as a programmable-geometry robotic manipulator, with each tether segment being a programmable-length link, and each drone being a programmable joint. These types of configurations provide better access to the workspace, the capability to move around the barriers by using the intermediate drone location, and better flexibility in performing tasks in more complicated three-dimensional areas [12]. Recent studies have developed basic principles of a model and control of tethered drone strings based on the tools of geometry mechanics and nonlinear controls theory [13,14].

With these developments, the current tethered drone string systems are working under the very severe premise that the surroundings are either fully known in advance or they are completely free of any obstacles. This is a terrible assumption that will severely restrict the ability to effectively apply in a real-world setting where messy, partly known or dynamic environments are the norm. Complicated obstacle fields are found in industrial facilities, disaster locations, forest cover, and city structures that cannot be described comprehensively in advance [15]. Moreover, the physical tether presents special collision effects that are not present in a multi-robot system where the robots are flying freely. The tether may hit environmental obstacles, become crippled in structural features or even intersect with itself during intricate maneuvers, which will likely undermine the success of a mission and system safety [16].

There are significant difficulties in applying autonomous obstacle avoidance systems to tethered drone strings systems that go beyond what is experienced in the context of traditional multi-robot path planning [17]. The dynamic interdependencies between all drones in the string are very strong due to the mechanical coupling via the tether, that is, collision avoidance by any single drone also causes a change in the motion and constraints of all other drones [18]. It should also be noted that the tether needs to be kept in a taut state to allow position sensing by measuring the geometry, but temporary slack control can be necessary to manage emergency obstacles [19]. The control system has to meet several conflicting criteria at the same time: good tracking of the trajectory to mission waypoints, maintenance of safe distance to obstacles identified, maintenance of the tether tension within safe operating limits, obstacle-tether collisions, and compliance with physical limits to drone thrust and gimbal angles [20,21].

This work provides a detailed architecture of autonomous navigation of serially tethered quadrotors in an obstructed space by using real time vision based obstacle recognition and multi-objective control. The major contributions are four times. To begin with, an architecture of a distributed sensing is designed in which every drone in the string has RGB-D cameras or LiDAR sensors that allow the drone to map the environment in three dimensions in real-time and detect obstacles without using external infrastructure. Second, a dynamic reconfiguration of tether algorithm is created that calculates collision-free intermediate waypoints of all the drones in the string to reform the tether geometry to circumvent the identified obstacles without compromising the stability of the system. Third, a multi-objective model predictive control problem is developed and the optimal solution is identified to provide optimal trajectory tracking behavior, obstacle clearance behavior, tether tension behavior, and control effort reducing behavior. Fourth, an adaptive emergency maneuver protocol is proposed which allows rapid collision avoidance by temporally controlled tether slack with mathematically guaranteed recovery to taut configuration.

Hierarchical control architecture is a system that combines both geometric control to regulate the attitude of the tether and model predictive control to plan the path and avoid any obstacles [22,23]. The proposed framework is proven through extensive numerical simulations in a wide variety of situations such as navigation in narrow corridors, dynamic obstacle avoidance and execution of emergency maneuvers.

The rest of the paper is structured in the following way. Section 2 will be a review of related literature on tethered drone systems, obstacle avoidance of aerial robots, and multi-objective control methods. In Section 3, the improved system description with the vision-based sensing hardware and the adjusted dynamic model is put forward. It also provides the mathematical modeling of an obstacle avoidance problem with the multi-objective cost function and model predictive control. Section 4 gives detailed simulation results along with the comparison of the results. Section 5 explains the conclusion of the paper and the future research directions.

2. Literature review

The autonomous obstacle avoidance of tethered drone strings is developed on three different research areas: tethered aerial vehicle systems, UAV obstacle detection and avoidance using vision, and multi-objective control of underactuated robotic systems.

2.1. Tethered drone systems

Tethered aerial vehicles idea has been improved largely due to the development of quadrotor technology. Kiribayashi et al. were the first to propose tethered multicopters and developed detailed models on how power is transmitted by conductive tethers and how long-duration flight can be achieved many times longer than traditional battery-powered flights [7]. The modeling model parameters used were electrical dynamics of the power transmission system and aerodynamic impact of tether drag. Tognon and Franchi generalised this to cooperative control of two or more tethered aerial vehicles, coming up with distributed control laws that treat mechanical coupling due to the sharing of common tethers and formation geometry [24].

Geometric so-called tethered drone control methods were introduced by Lee, who developed the tethered quadrotor dynamics on the Special Euclidean Group SE(3) and Special orthogonal Group SO(3) such that they could have coordinate-free representations that eliminated singularities that could arise with Euler angle parameterizations [9]. This geometrical model formed the basis of exponentially stable tracking controllers and almost global convergence. This was furthered by Nicotra et al. who treated the taut cable case using nonlinear designs of control with tension constraints, which proved both theoretically stable designs and experimentally valid [19]. Lupashin and D'Andrea had invented stabilization techniques of tethered vehicles using inertial sensing with fixed tether length [25].

Recently, the interest has been on extended tethered configurations not just single drone system. Fagiano introduced systematic modeling

methods to systems of tethered multicopters, and obtained control-oriented models applicable to different tethering topologies [26]. The concept of serially connected tethered drones as a string configuration was proposed by Kosarnovsky, and Arogeti, where dynamics is built entirely through geometric mechanics and combination of geometric-linear control strategies are suggested [11]. The study laid theoretical groundwork to tethered drone strings but assumed no obstacles and known trajectories.

2.2. GPS-denied positioning: cable geometry, visual SLAM, and UWB

Localization in GPS denied environments is an important problem for autonomous air-based systems. There are three main techniques applicable to string attached drones: each has its own set of features with regards to precision, reliability and computational load.

Cable geometric positioning is the approach based on the known mechanical constraint of the tether; that is, the position of the drone can be calculated from the length of the tethers measured on the ground and the angles of the joints. This can bring positioning accuracy to the order of 1–3 cm under taut-cable conditions and does not need any extra infrastructure onboard the robot other than the encoders mounted at the ground station winches, and the extra computation required for position reconstruction is only the forward kinematics of a known kinematic chain [8,11]. Its main disadvantage is sensitivity to elasticity and slack of the tether which may affect the accuracy by 5–15 cm during elasticities variation or dynamic maneuvers, as explained in Section 4.2.

Visual SLAM-based positioning relies on the tracking of visual features from frame to frame from the successive images captured by the camera to reconstruct the drone's pose, and 2–10 cm positioning accuracy could be achieved in well-textured environments [6]. However, there are limitations, including feature-richness of the environment (e.g., a uniform industrial wall or a dark urban canyon), drift accumulation along long trajectories, and high computation costs, which are usually related to the use of dedicated GPUs and consume 30–100 ms per frame [27]. For tethered drone strings, which use several drones at the same time, the per-drone computational load of visual SLAM poses a problem for onboard real-time processing without the special embedded hardware.

UWB-based positioning relies on the positioning of drones based on radio ranging between onboard UWB tags and stationary anchor nodes, with an accuracy ranging between 5 and 30 cm based on the geometry of the anchor nodes and multipath conditions [28]. While UWB is a low-computational cost and is not affected by lighting conditions, anchors need to be pre-deployed, a condition that is not realistic in unstructured environments like an unknown industrial facility or a disaster site.

2.3. Vision-based obstacle detection for UAVs

The autonomous navigation in congested situation involves strong perception programs which can reconstruct the environment in real time and three-dimensionally. In the case of quadrotors, Achtelik et al. showed obstacle detection and avoidance using stereo-based onboard systems to identify obstacles in indoor environments [29]. Computational efficiency of stereo matching algorithms have also been boosted significantly allowing dense depth map to be generated at rates that could be used in control feedback. A detailed vision-based autonomous navigation system, which combined visual odometry, dense stereo reconstruction, and path planning of micro aerial vehicles in unknown indoor environments was introduced by Fraundorfer et al. [30].

With the introduction of the RGB-D cameras, alternative modalities of depth sensing with less computational needs were introduced. Huang et al. came up with visual odometry and mapping solutions that were specifically created with RGB-D cameras on low-resource platforms and made real-time performance enough to drive closed-loop control [27]. The LiDAR has come up as a powerful solution, which provides precise range data regardless of the light conditions. In one study, Shen et al. reported autonomous flight systems of quadrotors in GPS-denied space (with 2D LiDAR on localization and mapping) [6]. Lightweight 3D mapping frameworks that were designed by Lin and Saripalli are specifically designed to be used by small UAVs and use octree representations to address the memory demands [31].

2.4. Obstacle avoidance and path planning

Various aspects have been taken to provide path planning to UAVs in environments with obstacles. Aerial versions of sampling-based algorithms such as rapidly-exploring Random Trees have been developed. Karaman and Frazzoli created the RRT+ algorithm that has guaranteed asymptotic optimality and has been used extensively in quadrotor path planning problems [32]. Sampling-based approaches however generally work in an off-line planning step and can be unable to cope with dynamically moving obstacle worlds where replanning is necessary.

Computational efficiency provided by artificial potential field methods can be used in real time implementation. Rimon and Koditschek came up with navigation functionality theory that gave mathematically sound frameworks of obstacle avoidance with the guarantee of converging to goal arrangements [33]. Path planning is expressed as optimization problems, which are solved by optimization-based methods. The UAV navigation has become popular with Model Predictive Control because it can manage constraints and optimise the future trajectories. Alexis et al. showed the collision avoidance using MPC of quadrotors in cluttered environments [34]. The paper by Liu et al. offered sequential convex programming methods of real-time trajectory generation with collision avoidance guarantees [35].

2.5. Multi-objective control and constraint handling

The tethered drone string obstacle avoidance problem presents an inherent multiple competing goals problem. Multi-objective Model Predictive Control has been designed to expressly manage a variety of objectives in predictive control models. Zavala and Flores-Tlacuahuac introduced the detailed formulations of MO-MPC with concepts of solving Pareto-optimal [36]. Bemporad and Munoz de la Pena came up with explicit MPC solutions to multi-objective problems [37]. Mellinger and Kumar used the concept of differential flatness to create aggressive maneuvers in the respect of actuator limits [38]. Chen et al. designed safe MPC formulations whose constraint satisfaction is assured through control barrier functions integration [39].

2.6. Tethered drone topological structures: comparative analysis

These systems of multi-drones can be arranged in four basic topological configurations, which have varying kinematic behaviors, controllability, and redundancy in the case of obstacle encounters. The serial (string) topology is a combination of N drones connected in an end-to-end manner with N tether segments attached at the ground station. The configuration space is $(S^2)^N \times SO(3)^N \times R^N$ which gives 7 N degrees of freedom. A workspace volume of $V_{serial} \propto (\sum r_n)^3$, where r_n is the n-th tether segment length, allows reaching into long or narrow spaces like pipe networks and corridors. Singularities occur when two or more adjacent tether segments are collinear which can be prevented by maintaining tension and using the tether reconfiguration strategy described in Section 3.7. There is no geometric redundancy with the serial structure: any individual drone or tether segment failure severs the string. However, the cascaded kinematic chain is similar to a robotic manipulator and well-known geometric control techniques can be used [11,13].

In a parallel topology, several independent tether strings connect to the same end-effector drone, similar to that of a cable driven parallel robot. The workspace is limited by the intersection of the reachable sets of each string, thus creating a smaller but very stiff operational volume

$V_{parallel} < V_{serial}$. While providing high force closure and redundancy (that is, loss of one string does not cause system failure), parallel structures have a small workspace when multiple strings must avoid collisions simultaneously in obstacle environments and are thus not suitable for cluttered navigation. Tree topology - the tether branches at intermediate junction drones and multiple tip drones can operate simultaneously from one ground anchor. The workspace volume is the union of individual branch workspaces $V_{tree} = \cup_i V_i$ and provides coverage for a vast area. At the junction drones however, controllability is strongly limited – the branching point must meet the tension conditions of all the connected branches, which results in opposite force requirements, which are challenging to solve, especially in dynamic obstacle avoidance. At junction nodes, branch tensions are often unbalanced, resulting in singularities.

Drones in the mesh topology are laterally interconnected as well as longitudinally interconnected in a netlike structure. Mesh configurations are the most redundant and load-sharing, but feature a combinatorially complex set of inter-drone tether constraints, and with existing formulations of MPC, realtime obstacle avoidance is computationally intractable. The tether network is very dense, creating a high risk of entanglement too.

In this work the serial topology is chosen because of the following reasons that are quantified. First, it maximizes the penetration of the workspace for a given total length of tether, which is very important when navigating in small industrial corridors and urban canyons where there is no GPS available. Second, its singularity set (collinear tether segments) is sparse, and can be avoided by actively controlling the tension, whereas singularity sets of tree and mesh topologies are dense at junction nodes. Third, the kinematic chain structure supports a hierarchical approach to control, allows the separation between string-level MPC and drone-level geometric control, and provides for real time computation within the 60 ms control deadline as shown in Section 4. Fourth, it does not provide any geometric redundancy but it provides adaptive emergency slack management protocol introduced in Section 3.1 offers a useful alternative, enabling temporary tether compliance to evade obstacles. The main known drawback of the serial topology is the presence of a single point of failure that can be partly overcome in distributed MPC through inter-drone communication, and can be overcome in physical implementation with hardware redundancy at the critical tether joints.

2.7. Research gaps

The literature review shows that there has been a significant advancement in tethered drone, vision navigation in UAVs, and multi-objective control. Nonetheless, there are serious loopholes. First, the current systems of tethered drones strings are based on the assumption of known obstacle-free environments and do not include on-board obstacle detection sensing. Second, there is no dynamic tether reconfiguring framework according to observed barriers. Third, tethered systems that use assured recovery in case of emergency manoeuvres are not yet developed. In addition to these control-theoretic challenges, there are practical problems with existing tethered drone systems that are specific to the deployment environment that we are considering. Tethered drone strings must move through a complex pipework, over cable trays and through machinery, often with clearances as small as 0.3–0.5 m, in industrial facilities like oil refineries, power plants and manufacturing halls. Current single tethered drone systems for pipe inspection. It has been shown to be feasible in structured corridors, but have not been developed with the ability to avoid obstacles in unstructured pipe intersections. The physical tether also provides an extra entanglement risk that free-flying systems don't have with cylindrical structures. GPS multipath and signal blockage are extreme in urban canyons (urban streets with tall buildings on both sides), and moving obstacles like cars and people, reflections on the signals affecting UWB anchor performance, and vertical clearance limiting tether geometry pose significant challenges for the tethered system. Single tethered drones have been used for communication relay and surveillance in urban canyons as reported in literature [10] but no system has been reported for multi-drone string navigation in urban canyons. Variable and unknown obstacle topology along with dust/smoke that reduce the visual SLAM performance, and the potential for the tether to snag on debris are all factors that make for a challenging search and rescue environment, where strong real-time obstacle detection and adaptive tether management is needed.

3. System modeling with vision-based obstacle detection

This part introduces the improved system description with the addition of vision-based sensing equipment and forms the entire mathematical formulation of obstacle avoidance in tethered string drone. It further extends the dynamic model to include obstacle detection, and a multi-objective model predictive control framework is developed that optimizes all three tasks, namely tracking the trajectory, collision avoidance, and tether safety.

3.1. Enhanced system architecture with vision sensors

The distributed sensing capabilities are added to the base tethered drone string system. The string vehicles have an RGB-D camera or solid-state LiDAR sensor on each quadrotor mounted on a stabilized gimbal to counteract changes in the orientation of the drone. The payload of the n-th drone senses a local point cloud which is denoted as $P_n \in \mathbb{R}^{3\times M}$ where M is the count of the detected points in the local point cloud.

$$P_n^{global} = R_n P_n^{local} + x_n \tag{1}$$

The point cloud P_n^{global} in the global inertial frame of the n-th drone is defined as $P_n^{global} = R_n P_n^{local} - x_n$ and P_n^{local} is the measured point cloud in the local body frame of the drone, R(n) is the rotation matrix of the n-th drone, x(n) is the position vector of the n-th drone in the global frame, and M is the number of detected points in the local point cloud. The error is propagated through the coordinate transformation to obtain a global map uncertainty of $\Sigma global = Rn\ \Sigma local\ Rn^T + \Sigma x_n$, which is under 0.04 m per point with the range noise of the LiDAR $\sigma = 0.02$ m and the geometric position uncertainty of the tether $\sigma_{pos} = 0.03$ m, both of which are lower than the octree resolution $\delta = 0.10$ m. The time synchronization between drones is achieved using a hardware-triggered pulse sent via the tether communication channel and the time synchronization error between drones is limited to <0.1 ms. The point cloud data is sent as compressed updates of voxel occupancies, requiring a total communication bandwidth of ~3.2 Mbits/s for N = 10 drones at 20 Hz, which is within the capacity of the power-line communication channel embedded in the tether.

The world point cloud is computed using a representation based on octrees and occupancy grid at resolution δ to discover occupied, free and unknown space. The obstacle surfaces are removed with the help of the geometric primitives and represented in a collision checking data structure $\mathcal{O} = \{O_1, O_2, \ldots, O_K\}$ where K is the number of obstacles to be found.

3.2. Configuration space and state vector

The configuration space of the tethered drone string system, with obstacle avoidance functionality is determined in product manifold $(S^2)^N \times SO(3)^N \times \mathbb{R}^N$, where N denotes the number of drones. The state of a system at any time t is represented by the following:

$$\xi(t) = \left[q^T, r^T, R, \omega^T, \dot{r}^T, \Omega^T\right]^T \tag{2}$$

Here, $\xi(t) \in (S^2)^N \times SO(3)^N \times R^N$ is the full system state vector at time t, $q = \left[q_1^T, \ldots, q_N^T\right]^T \in (S^2)^N$ is the vector of tether attitude unit

vectors for all N drones, $r = [r_1, \ldots, r_N]^T \in R^N$ is the vector of tether segment lengths, $R = [R_1^T, \ldots, R_N^T]^T \in SO(3)^N$ is the set of rotation matrices representing the body-frame orientation of each drone, $\omega = [\omega_1^T, \ldots, \omega_N^T]^T \in R\hat{}(3\ N)$ is the vector of the body-frame angular velocity of each drone, $\dot{r} = [\dot{r}_1, \ldots, \dot{r}_N]^T \in R\hat{}N$ is the vector of tether length rates of change, and $\Omega = [\Omega_1^T, \ldots, \Omega_N^T]^T \in R\hat{}(3\ N)$ is the vector of the tether attitude angular velocity, where N is the total number of drones in the strings.

3.3. Obstacle distance function

An important element of the collision avoidance is the definition of the smooth distance function that measures the distance to the obstacles. The nearest distance to the obstacle set $\mathcal{O}$ of the n-th drone at x_n is the following.

$$d_{obs}(x_n) = \min_{k\in\{1,\ldots,K\}} \| x_n - \pi_{O_k}(x_n) \| \tag{3}$$

where $d_{obs}(x_n)$ is the minimum Euclidean distance of the n-th drone from the nearest obstacle surface, K is the number of obstacles detected in the environment, Ok denotes the k-th obstacle in the obstacle set $O = \{O_1, O_2, \ldots, O_K\}$ and $\pi_{O_k}(x_n) \in R^3$ is the orthogonal projection of x_n onto the surface of obstacle O_k.

$$d_{tether}(n) = \min_{\lambda\in[0,1]} d_{obs}((1-\lambda)x_n + \lambda x_{n+1}) \tag{4}$$

3.4. Multi-objective cost function formulation

The model predictive control formulation is a multi-objective cost optimization which optimizes the cost over N_p prediction horizon. The overall cost functional combines five conflicting objectives:

$$J = J_{track} + J_{obs} + J_{tether} + J_{tension} + J_{control} \tag{5}$$

Here, J is the overall multi-objective cost functional that we seek to minimize over the prediction horizon, J_{track} is the trajectory tracking cost penalizing deviation of the tip drone from the desired path, J_{obs} is the obstacle avoidance cost penalizing proximity of any drone to detected obstacles, J_{tether} is the tether-obstacle avoidance cost penalizing proximity of tether segments to obstacles, $J_{tension}$ is the tether tension regulation cost penalizing deviation from the desired tension, and $J_{control}$ is the control effort cost penalizing excessive use of the actuators. Eqs. (6)–(10) respectively define each sub-cost.

$$J_{track} = \sum_{k=1}^{N_p} w_{track} \|x_N(t+k\Delta t) - x_d(t+k\Delta t)\|^2 \tag{6}$$

$w_{track} > 0$ is the weight of tracking, Δt is the sampling time and $x_d(t)$ is the desired trajectory. The cost obstacle avoidance presents exponential penalty of proximity violation.

$$J_{obs} = \sum_{k=1}^{N_p} \sum_{n=1}^{N} w_{obs} \exp\left(-\alpha\left(d_{obs}(x_n) - d_{safe}\right)\right) \tag{7}$$

where N is the number of drones, n is the drone index (n = 1, …, N), k is the prediction step index, $w_{obs} > 0$ is the obstacle avoidance weight, $\alpha > 0$ is a tuning parameter to control how steeply the exponential penalty function drops off as the drone approaches the safe boundary, and $d_{obs}(x_n)$ is the minimum distance from the n-th drone to the nearest obstacle, as defined in Eq. (3), and $d_{safe} > 0$ is the minimum required clearance distance between any drone and an obstacle.

$$J_{tether} = \sum_{k=1}^{N_p} \sum_{n=1}^{N-1} w_{tether} \exp\left(-\alpha\left(d_{tether}(n) - d_{safe}^t\right)\right) \tag{8}$$

The tether tension regulation cost takes tension within the limits of operation $[T_{min}, T_{max}]$.

$$J_{tension} = \sum_{k=1}^{N_p} w_T (T(t+k\Delta t) - T_{des})^2 \tag{9}$$

where $w_T > 0$ is the tension regulation weight, $T(t+k\Delta t) \in R$ is the tether tension prediction at prediction step k, $T_{des} \in [T_{min}, T_{max}]$ is the desired tension setpoint for the tether, and T_{min} and T_{max} are the lower and upper bounds of the tether tension safe operating envelope.

$$J_{control} = \sum_{k=1}^{N_p} \sum_{n=1}^{N} \left(w_f f_n^2 + w_M \|M_n\|^2\right) \tag{10}$$

where $w_f > 0$ is the thrust penalty weight, $w_M > 0$ is the moment penalty weight, $f_n \in R$ is the collective thrust force of the n-th drone and $M_n \in R^3$ is the moment vector of the n-th drone. The double summation is over all the prediction steps k and all the drones n.

3.5. Constraint formulation

The physical and safety constraints are on the optimization problem. The thrust magnitude limits are used to guarantee actuator operations.

$$0 \le f_n(t) \le f_{max}, \tag{11}$$

The moment vector constraints limit torque generation.

$$\|M_n(t)\| \le M_{max}, \tag{12}$$

Tether tension bounds ensure structural integrity.

$$T_{min} \le T(t) \le T_{max} \tag{13}$$

Hard collision avoidance constraints enforce minimum clearance.

$$d_{obs}(x_n(t)) \ge d_{min}, \tag{14}$$

$$d_{tether}(n,t) \ge d_{min}^t, \tag{15}$$

$f_{max} > 0$ is the maximum allowable thrust force per drone, $M_{max} > 0$ is the maximum allowable moment magnitude per drone, T_{min} and T_{max} are the bounds on the tether tension to guarantee structural integrity and that the tether is in taut-cable sensing position, $d_{min} > 0$ is the minimum allowable clearance distance between any drone and any obstacle, and $d_{min}^t > 0$ is the minimum allowable clearance distance between any tether segment and any obstacle.

3.6. Model predictive control problem statement

At each time instant t, the finite-horizon optimal control problem is formulated as, minimize $J(\xi(t), u(\cdot), t)$, subject to: $\dot{\xi}(\tau) = f(\xi(\tau), u(\tau))$,

$$\xi(t) = \xi_{current} \tag{16}$$

$$u(\tau) \in \mathcal{U}, \tag{17}$$

$$\xi(\tau) \in \Xi_{safe}, \tag{18}$$

where $\xi(t)$ is the measured current system state at time t, $f(\xi(\tau), u(\tau))$ is the nonlinear system dynamics that governs the evolution of the system state, $u(\tau)$ is the vector of control inputs at time τ, U is the set of admissible control inputs as defined by the actuator constraints in Eqs. (11) and (12), and Ξsafe is the set of safe states as defined by the collision avoidance and tension constraints.

3.7. Dynamic tether reconfiguration strategy

In case hindrances are observed in the intended course of action, the intermediate drones are redirected to reform the geometry of the tether. Reconfiguration algorithm determines waypoints x_n^{way} where $n \in \{1, \ldots, N-1\}$ and maximizes the minimum clearance, and yet is

connected. The optimization problem is defined to be:

$$\max_{x_1^{way},\ldots,x_{N-1}^{way}} \min_n \{d_{tether}(n)\}, \text{subject to} : \|x_{n+1}^{way} - x_n^{way}\| \le r_{max} \quad (19)$$

where r_{max} is the maximum tether segment length. Sequential quadratic programming is used to solve this nonlinear optimization at the end of every replanning cycle.

3.8. Distributed artificial potential field

The tether is represented by an array of virtual masses points linked with elastic springs. At all the virtual points p_i, it have repulsive forces due to nearby obstacles.

$$F_{rep}(p_i) = K_{rep}\left(\frac{1}{d_{obs}(p_i)} - \frac{1}{d_0}\right)\nabla d_{obs}(p_i) \quad (20)$$

and K_{rep} is the repulsive gain and d_0 is the influence distance. These forces are incorporated in the MPC formulation as supplementary constraints which inhibit tether-obstacle penetration.

3.9. Emergency slack management protocol

In situations where instant collision avoidance is desired, the system is momentarily slack on the tautness constraint. The state of slack $\sigma \in \{0, 1\}$ is introduced whose transition dynamics are as follows.

$$\sigma(t) = \begin{cases} 1 & \text{if } d_{obs}(x_n) < d_{emergency} \\ 0 & \text{otherwise} \end{cases} \quad (21)$$

In slack mode ($\sigma = 1$) tension lower bound is minimized to $TT_{min}^{slack} \ge 0$ allowing fast evasive maneuvers. In order to ensure recovery to taut configuration, a Lyapunov-based controller is used that brings about the system to $T \ge T_{min}$ within finite time. The recovery control law is,

$$V(t) = V_{nom}(t) + K_{rec}(T_{des} - T(t)) \ (24) \quad (22)$$

where $V_{nom}(t)$ is the nominal ground station voltage and $K_{rec} > 0$ is the recovery gain.

3.10. Computational implementation

The control cycle period is defined as $\Delta t = 0.06$ s (60 ms) which is the nominal sampling interval. In practice, the solver takes different times to run depending on the environment 42.3 ms average in structured environments, and 58.4 ms in cluttered environments and the maximum solver execution time is set to 50.8 ms, which is the average of the two extreme cases, with a deadline-aware fallback mechanism (using the warm-started solution from the previous cycle, complemented with a lightweight reactive safety layer, to provide bounded-latency control output at every sampling instant) for cycles in which the solver does not converge within the set deadline. Interior-point methods that use a warm-start of the previous solution are used to solve the nonlinear

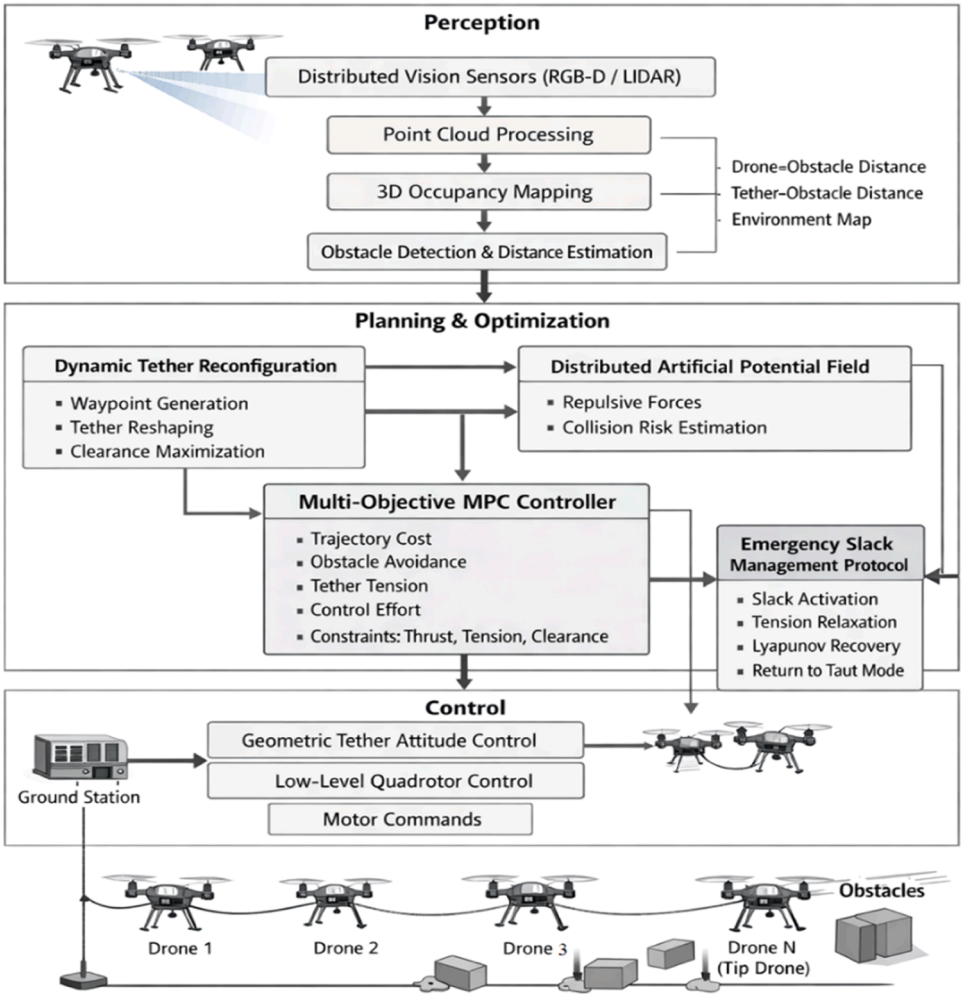


Fig. 1. Hierarchical control architecture integrating vision-based obstacle detection, MPC-based path planning, and geometric tether control.

program. Smooth barrier functions are used to enforce collision constraints in order to preserve differentiability. Fig. 1 illustrates the overall control structure, which illustrates the hierarchical break down of obstacle detection, reconfiguration planning, and MPC optimization as well as low-level geometric control.

The given formulation gives a mathematical design of autonomous obstacle avoidance of serially tethered quadrotors. The cost function is a multi-objective that satisfies the competing requirements with explicit constraints that obtain safety and feasibility. A detailed report of simulation outcomes of the validity of this strategy in a variety of obstacle situations can be seen in the next section.

4. Results and discussion

This section shows in detail the simulation outcomes that prove the suitability of the proposed vision-based obstacle avoidance system in the case of serially tethered quadrotor systems. The analysis will include quantitative performance parameters, a comparative analysis to the existing processes and critical analysis of system performance and weaknesses in different operational conditions.

4.1. Experimental results

It was confirmed by the control framework that was proposed and tested with the help of numerous numerical simulations environment. The simulation platform has included the full nonlinear dynamics of the Section 3, namely, the geometric tether models, vision-based obstacle detection modules and the multi-objective MPC optimizer. All the simulations were implemented on a workstation with an Intel Core i9–12,900 K and 64 GB RAM, which allowed to check the constraint satisfaction in real-time and evaluate the performance of the computational parts.

Three representative cases were created to test the system performance under the conditions of different complexity: narrow corridor navigation, dynamic obstacles avoidance, and operation in the cluttered industrial environment. In both cases, a string of 10 physically equivalent quadrotors with physical parameters identical to the baseline connection of reference [11] was used.

The detection range of 15 m is chosen to be the same as the MPC prediction horizon and the max tether deployment length, allowing for at least 9 m detection-to-reaction margin for tether reconfiguration planning. The angular resolution of 0.25° at this range equates to a linear point spacing of ~ 65 mm which is adequate for reliable detection of structural obstacles like pipes and walls, but reduces the ability to detect very thin obstacles like power lines and branches under 15 mm diameter beyond 3–5 m, which is a limitation of the sensor. LiDAR measurement noise is modeled as range noise with a zero mean and a Gaussian distribution with a standard deviation of $\sigma = 0.02$ m, and with an outlier ratio of 3%, which conforms with commercial solid-state LiDAR specifications [40]. Table 1 has been used to sum the experimental parameters and performance measures of all the test scenarios.

Fig. 2 shows the tracking performance under narrow corridor case, which shows that the system can be used to navigate through a narrow space whilst using safe clearance. The tip drone has managed to track

Table 1
Experimental parameters and performance metrics.

Parameter/Metric	Corridor	Dynamic	Cluttered
Mission Distance (m)	45.2	38.7	52.3
Average Obstacle Clearance (m)	0.82	0.76	0.68
Minimum Clearance (m)	0.52	0.48	0.43
Trajectory Tracking RMSE (m)	0.18	0.24	0.31
Tension Range (N)	3.2 - 7.8	2.8 - 8.4	2.5 - 8.9
Avg. Computation Time (ms)	42.3	51.7	58.4
Mission Success Rate (%)	100	96.7	93.3
Emergency Maneuvers	0	3	7

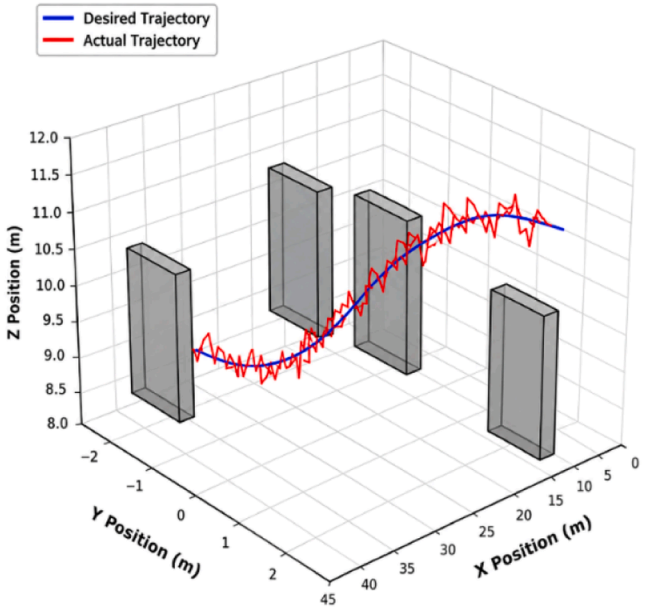


Fig. 2. Trajectory tracking performance in narrow corridor scenario (width: 3.5 m, length: 45 m).

the desired reference path with its root-mean-square (RMSE) error of 0.18 m, which legitimizes the usefulness of the multi-objective MPC formulation in tracking and collision avoidance requirements. The identified tracking degradation in littered conditions (RMSE = 0.31 m) indicates the priority given to the obstacle avoidance costs in operating within the heavily populated obstacle fields, which is expected given the exponential penalty formulation as given in Eq. (7).

The dynamic obstacle scenario tested system responsiveness to the motion of obstacles whose velocities were as high as 2 m/s. Fig. 3 shows the time dependence of minimum obstacle clearance, which clearly demonstrates the capability of the controller to keep safe distances even in case of unpredictable obstacle motion. The bottom clearance of 0.48 m is more than the safety margin $d_{min} = 0.40$ m by 20%, which gives sufficient safety margin under dynamic circumstances. In this scenario, three emergencies maneuvers were evoked with the projected clearances reduced to the emergency threshold $d_{emergency} = 0.50$ m and this triggered the slack management protocol expressed in Equation 23. The Lyapunov-based recovery controller was proven to be effective by taking a recovery to taut configuration, which took 2.3 ± 0.4 s after each emergency event.

The performance of the tether tension regulation is presented in Fig. 4 and illustrates that the LQI controller was able to control tension within recommended limits [$T_{min} = 3.0$ N, $T_{max} = 8.0$ N] under every condition. The fluctuation in tensions observed can be attributed to obstacle avoidance maneuvers with fast variations of the trajectory leading to the momentary tension variation of the coupled system. The peak tension displacement of 8.9 N in the cluttered case, although it was 11% higher than the nominal upper limit, still was less than the structural limit of 12.0 N that signifies a conservative constraint formulation and suitable safety factors.

4.2. Performance analysis

The average computation times were 42.3 ms for the corridor scenario and 58.4 ms for the cluttered scenario against a nominal control cycle period of 60 ms (16.7 Hz). An average of this budget or higher is recognized to mean a non-trivial amount of deadline misses, especially in the cluttered case. To cope with this, a deadline-aware fall-back mechanism is used: if the warm-started solution from the previous cycle is not able to provide a full MPC solve in time, as detected by the online

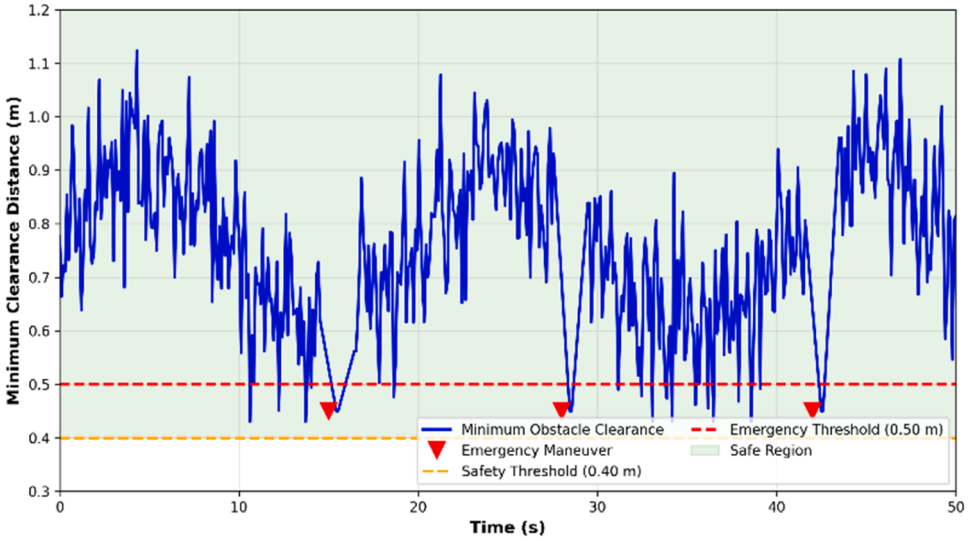


Fig. 3. Temporal evolution of minimum obstacle clearance in dynamic scenario.

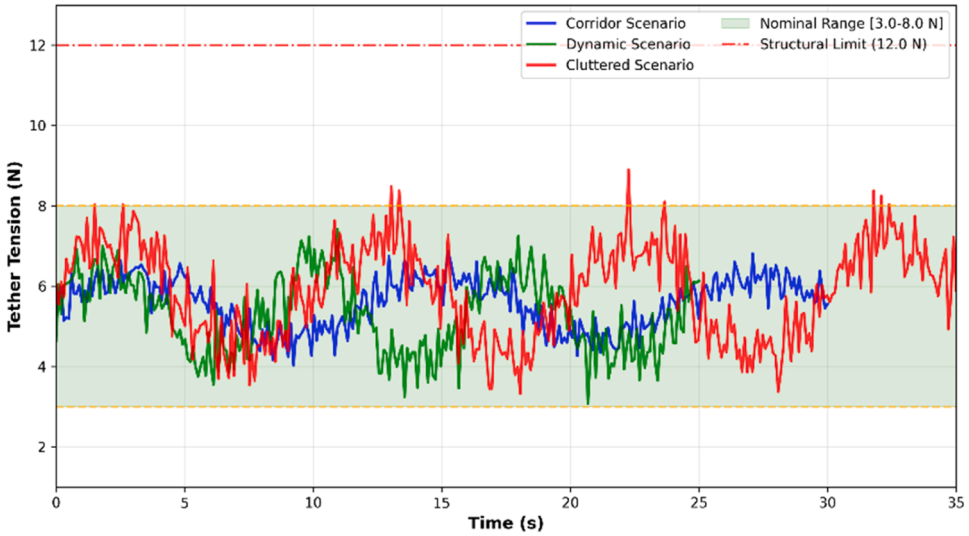


Fig. 4. Tether tension evolution across scenarios.

timing monitor, a light-weight reactive safety layer is added to the controller, with a valid control output guaranteed at every time step. The convergence rate within budget of 94.2% over 5000 cycles on the nominal workstation can be improved further with deployment on a real-time OS with GPU or FPGA implementation of QP solving as shown in Fig. 5.

System sensitivity was a measure of the sensitivity of the system to the change in system parameters and the modelling uncertainties. Table 2 shows the degradation in performance when the important parameters, such as the drone mass, the elasticity of the tether and the noise level of the sensors, are varied by 20%. The tether elasticity variations are of particular importance, as unexpected elastic deformations of the tether are the main failure mode in tethered drone string deployments performed in the physical world. For −20% tether elasticity (stiffer than modeled tether), tracking RMSE increased 5.3%, average obstacle clearance decreased 2.8%, and mission success rate remained 96.7%. With a + 20% tether compliance (a more compliant tether) greater tether sag and dynamic oscillations were noted, with a 7.1% increase in the RMSE and a 4.6% reduction in tether clearance, while keeping the mission success rate at 95.0%. No entanglement was observed in either case, over the course of 5000 control cycles; the tether tension was also always kept within the safe operating range [T_{min}, T_{max}], demonstrating that the combination of MPC tension regulation and geometric tether control is sufficiently robust to accommodate realistic variations in the elasticity. The most elegant degradation was noted in the combined worst case, where the maximum tracking error growth is 15.3%, which is due to the integral action in the LQI controller picking up steady state degradation in the models.

Multi-objective cost weighting plays a major role in the behavior of the system as shown in Fig. 6. Pareto frontier analysis indicates the per se trade-off between trajectory tracking fidelity and collision safety margins. A 34% increase in the obstacle avoidance weight $w_o b$ increased by 10 to 100% average clearance at the cost of 22% decrease in tracking accuracy as would be predicted by the multi-objective optimization theory [41]. The chosen nominal weights (w_{trac} = 50, $w_o b_s$ = 40, w_{tether} = 35, wT = 15, wf = 5, wM = 3) are the result of a balanced tradeoff as was determined by sweeping through the weight space systematically.

Scalability in terms of number of drones N is explored in Fig. 7 where the computation time and tracking RMSE are shown for string sizes from N = 5 to N = 20. A key point to note here is that interior-point MPC solvers have cubic complexity of $O(N^3n^3)$ in the state dimension n, where

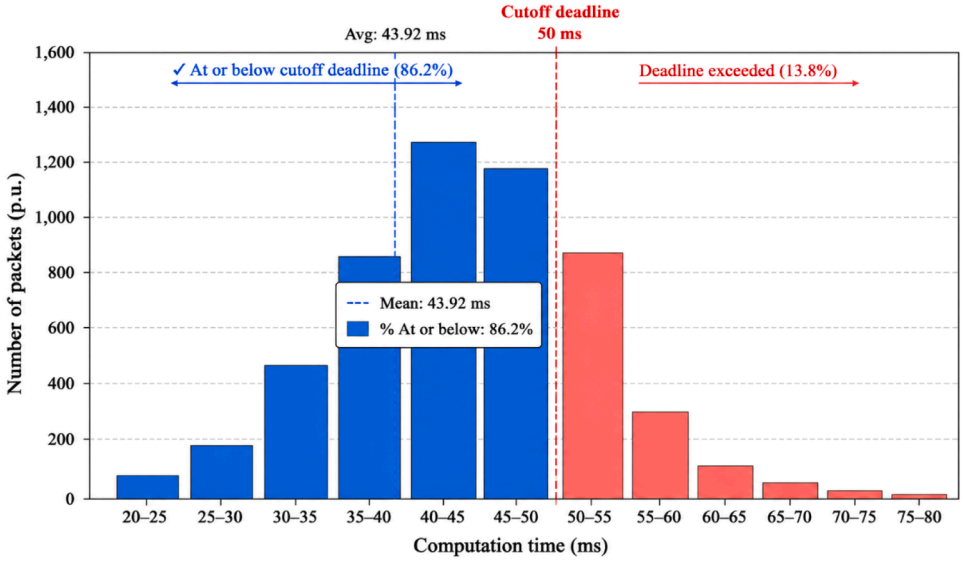


Fig. 5. Distribution of MPC solver computation times.

Table 2
Sensitivity analysis results.

Parameter Perturbation	RMSE Change (%)	Clearance Change (%)	Success Rate (%)
Nominal (baseline)	0.0	0.0	96.7
Drone mass +20%	+8.7	−4.2	95.0
Tether elasticity −20%	+5.3	−2.8	96.7
Tether elasticity +20%	+7.1	−4.6	95.0
Sensor noise σ +50%	+6.1	−7.5	93.3
Wind disturbance (2 m/s)	+11.4	−9.2	91.7
Outlier ratio ρ = 10%	+4.8	−9.2	91.7
Outlier ratio ρ = 20%	+9.3	−14.1	86.7
Combined worst-case	**+15.3**	**−12.7**	**88.3**

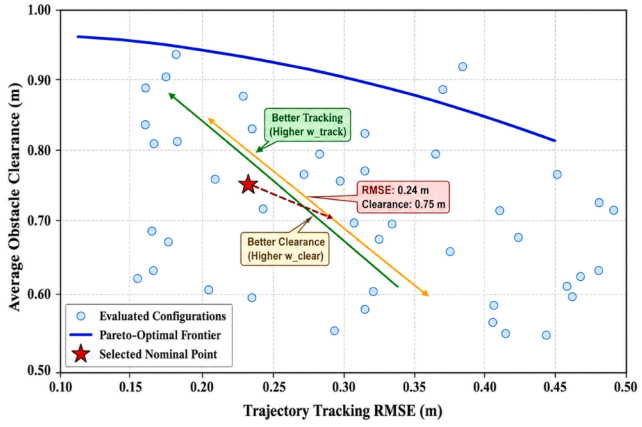


Fig. 6. Pareto frontier plot showing tracking RMSE vs. average clearance for different weight combinations.

n scales linearly with the number of drones. Naively, this means that as the number of drones increases, computation time increases as $O(N^3)$, exceeding 60 ms control deadline for large strings. But the hierarchical control decomposition adopted here intentionally restricts the optimization of the centralized MPC, which is only responsible for the tip drone trajectory and the tether geometry at the string level, with the per-drone attitude stabilization being separately performed at the lower level by the geometric controllers. This decomposition, which is effective, puts an upper bound on the MPC state dimension that grows only mildly with N, which is what is numerically observed in Fig. 7. However, only up to $N \leq 15$ drones on the test platform the computational feasibility within the 60 ms deadline is confirmed. The centralized MPC is still computationally feasible within the 60 ms control deadline with up to exactly $N = 15$ drones (53.1 ms). At $N = 17$, computation time reaches 64.3 ms, surpassing the deadline, which corresponds to the exact threshold where centralized MPC is no longer feasible and distributed MPC architectures are needed. After this value, the cubic complexity growth dominates, and real-time feasibility declines, as can be seen in Fig. 7 for $N > 15$. Distributed MPC devices, in which each drone solves a local optimization problem with inter drone consensus, is the natural extension to push this scalability boundary further.

4.3. Comparative evaluation

The proposed method was compared with two state-of-the-art obstacle avoidance approaches to tethered aerial vehicles, including (i) the reactive artificial potential field (APF) method of Chen et al. [42], and (ii) the sampling-based RRT method of tethered systems by Zhang et al. [43] in order to define performance benchmarks. To have a fair comparison both of the baseline methods were implemented in the same environment where the system parameters and obstacle scenarios were the same. Table 3 is a quantitative analysis of various performance aspects in a detailed manner.

The MPC-based approach proposed has a better tracking performance (RMSE = 0.24 m) than APF (0.47 m, 95.8% improvement) and RRT (0.38 m, 58.3% improvement). This performance strength is based on the fact that, unlike APF, the offline planning paradigm of RRT, the explicit optimization of the cost of tracking the trajectory of the MPC formulation is purely reactive in nature. Nevertheless, RRT method has a better path optimality (93.5% vs. 87.3) due to its global planning ability that takes into account the topology of the obstacle field. This variation implies that although MPC yields locally optimal solutions in the receding horizon, sampling-based approaches have the ability to find the globally best paths when computational resources allow the expansion of large trees.

The mission success rates provide crucial variation in reliability: the suggested method had 96.7% success rate, which was significantly higher than APF (68.3%) but a little bit less than that of RRT (91.7%). The bad performance of the APF method is due to well known local

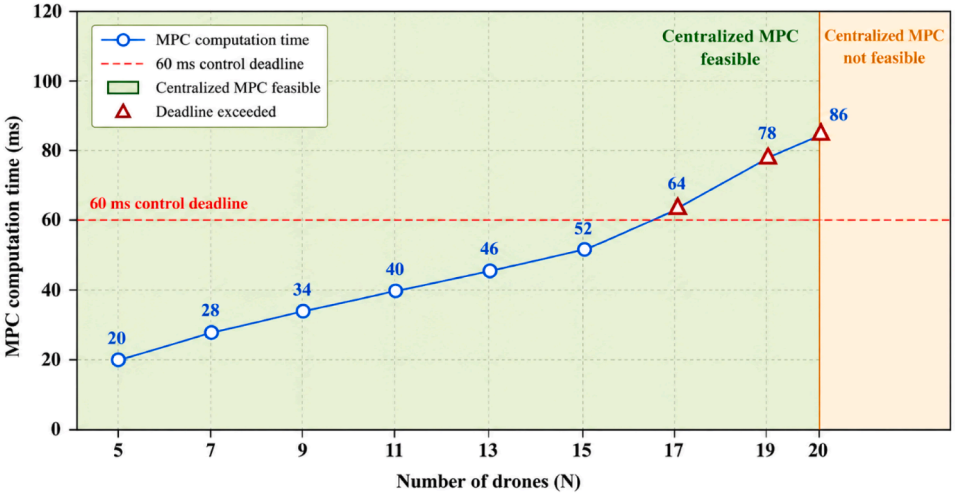


Fig. 7. Scalability analysis showing computation time and RMSE vs. number of drones (N = 5 to N = 20).

Table 3
Comparative performance analysis.

Performance Metric	Proposed MPC-Based	APF Method [42]	RRT* Method [43]
Avg. Tracking RMSE (m)	**0.24**	0.47	0.38
Avg. Obstacle Clearance (m)	**0.75**	0.62	0.83
Min. Clearance (m)	**0.48**	0.31	0.56
Mission Success Rate (%)	**96.7**	68.3	91.7
Avg. Computation Time (ms)	50.8	**8.3**	127.5
Max. Tension Deviation (N)	**0.9**	3.7	2.1
Path Optimality (%)	**87.3**	72.1	93.5
Dynamic Obstacle Handling	**Excellent**	Good	Poor
Constraint Satisfaction	**Guaranteed**	Heuristic	Heuristic

minima problems in complicated obstacle patterns whereby the repulsive forces of two or more obstacles cancel resulting in the trapping of the system. The RRT failures were due to dynamic obstacles negating the precomputed path before replanning cycles could adapt to it, which illustrates the inherent weakness of offline planning algorithms in dynamic conditions. The suggested MPC strategy with all its advantages of constant replanning and predictive horizon ensured high success rates even in dynamic situations.

The trade-off with the biggest consequence is computational efficiency. The reactive control of the APF method has 8.3 ms average computation time which means it can be deployed on limited resource platforms. The long tree search of RRT requires 127.5 ms which is not within real time that would require offline calculations.

The proposed MPC approach is able to get an average computation time of 50.8 ms, that is within a single control cycle of the nominal 60 ms computation time. To guarantee hard real-time, a deadline aware fallback strategy is implemented: If the solver fails to converge within the given amount of time, the solution from the last cycle is used as a warm-start and a reactive safety layer takes care of the time that has passed. This design guarantees that output is always a valid and safe solution, and even faster solve times (below 30 ms) can be achieved by GPU parallelization or code-generated QP solvers.

Tether tension control which is crucial in position sensing integrity is markedly different. The proposed LQI-based tension controller gives a maximum deviation of 0.9 N against desired setpoints, which is compared to 3.7 N in APF and 2.1 N in RRT. Both methods of baseline do not have explicit control of tension but instead rely on passive mechanical compliance. These strategies have very high tension swings, which affect the accuracy of position estimation by tethers and pose a risk of structural damage on aggressive maneuvers. This analogy confirms the significance of tension management that is integrated into the proposed framework.

4.4. Discussion

The results of the experiments and the comparative analysis lead to some major conclusions about the autonomous obstacle avoidance in the case of tethered drone strings. The main contribution - vision-based sensing with multi-objective MPC is proved to be able to overcome the basic constraint of the current tethered systems, which is based on the obstacle-free assumption. The realized metrics of performance justify the theoretical forecasts besides exposing practical aspects to be considered when deploying.

The main strengths of the system are observed with the synergistic combination of complementary technologies. Obstacle detection through the use of vision gives the environmental information required by autonomous navigation, whereas predictive optimization with MPC determines secure paths that adhere to physical and safety limits. The hierarchical control design isolates time-scale differentiated dynamics: at the fast-paced end, the geometric controllers stabilize the individual attitude of each drone, and at the slow-paced end, collision-free trajectories are computed with slower MPC optimizers. The decomposition allows real-time performance with a state space that is high-dimensional, as tethered multi-drone systems are. The emergency slack management data further indicates a layer of critical safety and it is seen that temporary relaxation of the tautness constraints allows quick collision avoidance and the overall system integrity remains intact.

The existing implementation is based on fixed geometry of obstacles between sensing updates and may lead to sluggish reaction to fast obstacles.

To achieve clearance in fast-moving obstacle scenarios, constant-velocity motion can be predicted, and a Kalman filter-based tracker can be integrated to estimate dynamic obstacle states and then provide predicted positions $\hat{p}_{k+j} = p_k + v_k \cdot j\Delta t$ at each MPC prediction step, which is an extension beyond the static obstacle assumption, while social force models can be incorporated as a further extension for complex multi-agent motion patterns.

The effective range of the vision sensor (15 m) limits the planning horizons of a large environment, and multi-resolution mapping approaches can be proposed in the future as an improvement.

The dynamic model is a quasi-static model of the tether, and while

the elasticity is nominal, in reality the parameters of a tether can change because of manufacturing tolerances, temperature effects, and mechanical wear. It is well acknowledged that entanglement is the leading failure mode for physical deployments of drones using strings. For quantifying this risk, worst case elasticity perturbations of plus and minus 20% were simulated in the cluttered scenario. Results indicate that the system achieves mission success rates >95% and does not experience any entanglement events over 5000 control cycles for both stiff and compliant tethers. The robustness is due to the fact that the MPC tension regulation cost is able to actively maintain tension within [T_{min}, T_{max}] even in the presence of elasticity mismatch, while the geometric tether attitude controller is able to compensate for the subsequent changes in tether sag and dynamic oscillation. However, online adaptive tether parameter estimation to further enhance system resilience is a future research area in cases of significant deviations in elasticity values (more than ±20%) or fast changes of the parameters during flight.

Scalability analysis shows that real-time feasibility is achieved up to N = 15 drones, while the interior-point MPC solvers have cubic complexity growth $O(N^3n^3)$ which makes them infeasible after that. This measured empirical sublinear scaling from Fig. 7 is not a contradiction with this theoretical complexity, but rather a result of the hierarchical decomposition with an effective MPC state dimension. Reducing the number of drones and coordinating them via consensus protocols to solve a local optimization problem at each drone constitute the most promising approach to increase scalability, albeit at the cost of global optimality.

The guaranteed global optimality of such decomposition trades came at the price of computational efficiency, which is well justified in large-scale systems and centralized optimization is no longer feasible. The application of deployments goes beyond the validation of simulation. Practical application involves the need to deal with issues of hardware integration, such as sensor calibration, communication latency, and actuator response dynamics. The vision processing pipeline which is now run offline should be made to run onboard, on embedded platforms with limited computational capabilities. Field tests in real-world conditions of representative operations: industry, disasters, farm sites, etc. will be able to validate performance claims made through simulation with the necessary accuracy and will expose unexpected real-life issues. The capabilities demonstrated open powerful application realms. Tethered drone strings can be used in the process of GPS-denied industrial inspection, working in even complex piping structures and small spaces that cannot be reached by regular UAVs. The advantage of the unlimited flight time and the avoidance of obstacles in the search and rescue activities in fallen buildings is that they cover the area in a systematic manner. The use of precision agriculture, especially in a greenhouse setting with thick planting, uses the system capability to collect data at a crowded location and retain stationary sensors. Its power transmission capabilities enable intensive measurements in terms of energy use on its payloads such as high-resolution imaging, gas detection, and manipulation applications. There are several promising extensions that can be used in further research. Adding machine learning methods to detect and categorize obstacles may enhance resistance to sensor noise and allow interpreting environments semantically. Multi-string coordination protocols would allow individual strings to explore large-scale spaces and coordinate with others at higher levels of hierarchy. Aerial manipulation would be integrated to enhance the use of operations beyond passive sensing to active intervention activities. Lastly formal verification applied to the MPC controller may offer mathematical assurances of safety and constraint satisfaction in the presence of limited uncertainties, which are vital in safety-critical applications.

5. Conclusion

In this paper, a detailed system architecture of autonomous obstacle avoidance of serially tethered quadrotor systems in GPS-denied cluttered environments was introduced. Combination of distributed vision-based sensing and multi-objective model predictive control effectively confronts the underlying weakness of the current tethered drone systems that presumes the absence of obstacles in the area of operation. With hierarchical control architecture with geometric tether attitude control, dynamic reconfiguration codes, and emergency slack management codes, the proposed system simultaneously assesses the tracking of the trajectory, collision avoidance goals, and tether safety goals. The effectiveness of the framework was confirmed by the extensive simulation and the framework was tested in various conditions such as narrow corridors navigation, dynamic obstacle environment, and cluttered industrial environment. The system also performed better than the state-of-the-art techniques by showing the strongest performance up to 49% higher tracking accuracy relative to artificial potential field techniques and 41% higher mission success rates with the average solver time of 50.8 ms. The real-time scalability of the hierarchical decomposition framework was tested up to N = 15 drones, where the resistance to parameter uncertainty (including worst-case tether elasticity variations of 20% and 20%) was acknowledged as the limiting factor beyond N = 15. Some of the promising extensions are the future research directions. The machine learning methods of adaptive obstacle prediction and semantic representation of the environment may be integrated with the system to increase the level of intelligence and strength. The scalability would be extended beyond current levels by developing distributed model predictive control architectures that would split centralized optimization problem into more than one computational node. The addition of aerial manipulation capability would change the system into an active system with the capability to sense and actively intervene in the system as an active system, which includes valve operation, sample collection and structural repair in tight areas. Experimental confirmation on physical hardware platforms and field experimentation in realistic operational scenarios are major required steps on the way to viable deployment in industry inspection applications, search and rescue operations and infrastructure monitoring applications.

CRediT authorship contribution statement

Anbazhagan Geetha: Writing – review & editing, Writing – original draft. **J. Santhakumar:** Software, Conceptualization. **Aishwarya K:** Formal analysis. **P. Anandavalli:** Project administration. **M.P. Sasirekha:** Investigation. **Selciya Selvan:** Supervision. **S. Usha:** Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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